

Table 1: BPMN at three levels of complexity

	Basic	Intermediate	Advanced
Activities	Abstract	(activity-human.png) , Service (service-activity.png) , Call (call-subprocess-activity.png)	
Events	Start, End	Message, timer, error, signal	Event sub-process
Gateways	Parallel, exclusive		
Sequence flow	Sequence		
Others	Pools, lanes, annotation, links		Loop, multi-instance

Generally, to create an application, the application pages, data model and processes are defined in parallel and evolve as you go deeper in the application details. The use case that we will be following is given below.

An employee opens the Travel Tool application to view pending and approved travel requests. He/she creates a new travel request. Then the request is sent to the employee's manager. The manager reviews the request and either approves or refuses it.

See <https://github.com/Bonitasoft-Community/getting-started-tutorial> for the full artifacts.

- Design application pages
- Create the data model
- Create the process definition
- Create forms
- Run process
- Build the application

Designing application pages

The first step in creating an application is to design the application pages in the UI designer using dummy data, creating a prototype application as seen by users. The Bonita Studio UI designer is an environment for creating application pages and forms. To start the UI designer, click its icon in the Bonita Studio Cool bar (at the top of the screen). The UI designer opens in a browser.

Creating the data model

There are two stages to defining variables — first define the business data model, and then specify how the process handles objects of the model. The model is defined as a set of Java classes, and the process uses instances of these classes.

Creating the process definition

Creating the diagram: The first stage is to create the new diagram, which you do using Bonita Studio.

Create the diagram as follows:

1. Click *New diagram* on the Bonita Studio *Welcome* page. This creates an almost empty diagram for you to start updating.
 - The large rectangle with a name at the left is the pool.
 - Inside the pool there is a lane, which is also a rectangle. You can see the border of the lane at the left side, beside the pool name. The other borders of the lane coincide with the pool border and so are not visible.
 - The green circle in the lane is a start event.
 - The blue box in the lane is a human task.
2. The first thing to do is to give the diagram a more descriptive name. Click on the diagram outside the

pool, then go to the *Details* panel.

This is the area on the bottom-right of the screen.

3. In the *General* tab, in the *Diagram* pane, click *Edit* next to the *Name* field.
 4. Enter the new diagram's name, *Simple Travel Request*, in the popup and click *OK*. You can see that in the tab at the top of the whiteboard, the diagram name has changed.
 5. Now give the pool a more descriptive name. Select the pool by clicking in the space between the pool border and the lane border at the left of the diagram. Then go to the *Details* panel, then the *General* tab, followed by the *Pool* pane, and click *Edit*, which is next to the *Name* field. Enter the new pool name, *Travel Request*, in the popup. When you have renamed the diagram and the pool, the diagram will look like what's shown in Figure 6.
 6. This process starts when an employee fills in a travel request form. You do not need to create a task for this because it is this action that triggers the process to start. This is known as process instantiation. A form for process instantiation is defined at the pool level.
 7. Define what happens after the user submits a request form — the manager reviews the travel request and approves or refuses it. You can use the example task that was added to the diagram automatically. Click on the task name and change it to *Manager review*.
 8. Later on in this process, the subsequent task would be to send the travel request details to the admin team so that they can handle the booking. For now though, we are just interested in getting the first part of the process working; so add an end after the review task. You can do this by dragging the end icon (red circle) from the palette to the whiteboard, and then connecting to the *Manager review* task with a flow element.
- When you have finished, your diagram should look like what's shown in Figure 7.